

Copilot Commands

Commands which we can use in Visual Studio, copilot chatbot- Agent mode:

<code>/explain</code>	Explains selected code in plain English
<code>/fix</code>	Suggests fixes for selected code
<code>/optimize</code>	Suggests performance or readability improvements
<code>/tests</code>	Generates unit tests for the selected code
<code>/doc</code>	Writes documentation comments
<code>/generate</code>	Creates new code (e.g. boilerplate, patterns)
<code>/review</code>	Reviews code for issues
<code>/commit</code>	Generates commit message for staged changes

One Example In Java in Visual Studio

Suppose you have one buggy method:

```
public int divide(int a, int b) {  
    return a / b;  
}
```

If you highlight this and type `/fix` in Copilot Chat → it will suggest adding a division-by-zero check.

`/tests` might generate a JUnit test class for it.

`/explain` will describe in plain language what the method does.

That code we can accept and reject

GitHub Copilot CLI Commands and VS Code Prompts (with Examples)

These are used in your terminal with the GitHub CLI extension (`gh copilot``):

Command: `gh copilot suggest "list all files modified in last 24 hours"`

Example(Given by copilot): `find . -type f -mtime -1`

Description: Finds recently modified files.

Command: `gh copilot suggest "search for TODO comments"`

Example: `grep -r "TODO" .`

Description: Searches for TODOs in code.

Command: `gh copilot suggest "create a new git branch"`

Example: `git checkout -b feature-login`

Description: Creates a new branch.

Command: `gh copilot suggest "create a @RestController for managing users"`

Example:

```
@RestController
```

```
@RequestMapping("/users")
```

```
public class UserController {
```

```
    @GetMapping
```

```
    public List<User> getAllUsers() {
```

```
        return List.of(new User("John"), new User("Jane"));
```

```
    }
```

```
}
```

Description: Generates a REST controller with a GET endpoint.

2. Create a Service Layer

Command: `gh copilot suggest "create a @Service class for User operations"`

Example:

@Service

```
public class UserService {  
    public List<User> getUsers() {  
        return List.of(new User("John"), new User("Jane"));  
    }  
}
```

Description: Business logic service for handling user operations.

Command: gh copilot suggest "check disk usage"

Example: df -h

Description: Shows disk space.

Command: gh copilot suggest "download file from URL"

Example: curl -O https://example.com/file.txt

Description: Downloads a file.

Command: gh copilot suggest "find all Java files"

Example: find . -name "*.java"

Description: Lists Java files.

Command: gh copilot suggest "kill process by name"

Example: pkill node

Description: Terminates a process.

Command: `gh copilot suggest "check system uptime"`

Example: `uptime`

Description: Shows how long system has been running.

Command: `gh copilot explain "git reset --hard HEAD"`

Example: "Resets working directory and index to last commit."

Description: Explains Git command.

Command: `gh copilot explain "chmod 755 script.sh"`

Example: "Gives owner full access, others read/execute."

Description: Explains file permissions.

Command: `gh copilot explain "docker run -it ubuntu bash"`

Example: "Runs Ubuntu container interactively."

Description: Explains Docker flags.

Command: `gh copilot config set execute=true`

Description: Enables command execution.

Command: `gh copilot --help`

Description: Shows help menu.

Copilot CLI Commands for Spring Boot Project

3. Create a Repository

Command: `gh copilot suggest "create a @Repository interface for User entity"`

Example:

@Repository

```
public interface UserRepository extends JpaRepository<User, Long> {  
    Optional<User> findByUsername(String username);  
}
```

Description: Spring Data JPA repository with a custom finder method.

4. Generate CRUD Endpoints

Command: gh copilot suggest "create CRUD endpoints in @RestController for User entity"

Example:

@PostMapping

```
public User createUser(@RequestBody User user) { return repo.save(user); }
```

@PutMapping("/{id}")

```
public User updateUser(@PathVariable Long id, @RequestBody User user) { ... }
```

@DeleteMapping("/{id}")

```
public void deleteUser(@PathVariable Long id) { repo.deleteById(id); }
```

Description: Auto-generates controller methods for CRUD.

5. Generate Unit Tests for Service

Command: gh copilot suggest "generate JUnit test cases for UserService"

Example:

@Test

```
void testGetUsers() {
```

```
    UserService service = new UserService();
```

```
    assertEquals(2, service.getUsers().size());
```

```
}
```

Description: Basic unit test for a service method.

6. Add Exception Handling

Command: gh copilot suggest "create a @ControllerAdvice for global exception handling"

Example:

```
@ControllerAdvice
```

```
public class GlobalExceptionHandler {  
  
    @ExceptionHandler(Exception.class)  
  
    public ResponseEntity<String> handleException(Exception ex) {  
  
        return ResponseEntity.status(HttpStatus.INTERNAL_SERVER_ERROR)  
            .body("Error: " + ex.getMessage());  
    }  
}
```

Description: Centralized exception handling.

7. Create DTO and Mapper

Command:

gh copilot suggest "create a UserDTO and mapper for User entity"

Example:

```
public record UserDTO(Long id, String username) { }  
  
public class UserMapper {  
  
    public static UserDTO toDTO(User user) {  
  
        return new UserDTO(user.getId(), user.getUsername());  
    }  
}
```

Description: Demonstrates DTO pattern with mapping.

8. Add Swagger/OpenAPI Config

Command:

gh copilot suggest "add Swagger configuration for Spring Boot project"

Example:

```
@Configuration
```

```
@OpenAPIDefinition(info = @Info(title="User API", version="1.0"))
```

```
public class SwaggerConfig { }
```

Description: Generates Swagger docs for REST APIs.

9. Generate Security Config

Command:

gh copilot suggest "create a Spring Security config with in-memory users"

Example:

```
@Configuration
```

```
public class SecurityConfig {
```

```
@Bean
```

```
public SecurityFilterChain security(HttpSecurity http) throws Exception {
```

```
    http.authorizeHttpRequests(auth -> auth.anyRequest().authenticated())
```

```
        .httpBasic();
```

```
    return http.build();
```

```
}
```

```
}
```

Description: Adds Spring Security basic authentication.

GitHub Copilot CLI Commands

These use the gh copilot extension in your terminal.

Command	Example Usage	What It Does
gh copilot suggest	gh copilot suggest "list all files modified in last commit"	Suggests: git diff --name-only HEAD~1
gh copilot suggest	gh copilot suggest "create a new branch"	Suggests: git checkout -b new-feature
gh copilot suggest	gh copilot suggest "delete all .log files"	Suggests: find . -name "*.log" -delete
gh copilot suggest	gh copilot suggest "check disk usage"	Suggests: du -sh *
gh copilot suggest	gh copilot suggest "find all TODOs in code"	Suggests: grep -r "TODO" .
gh copilot suggest	gh copilot suggest "install Node.js"	Suggests: sudo apt install nodejs
gh copilot suggest	gh copilot suggest "start a Python HTTP server"	Suggests: python3 -m http.server
gh copilot suggest	gh copilot suggest "undo last git commit"	Suggests: git reset --soft HEAD~1
gh copilot suggest	gh copilot suggest "show current IP address"	Suggests: ip addr show
gh copilot suggest	gh copilot suggest "compress a folder"	Suggests: tar -czvf archive.tar.gz folder/
gh copilot explain	gh copilot explain "git rebase -i HEAD~3"	Explains interactive rebase
gh copilot explain	gh copilot explain "chmod 755 script.sh"	Explains file permissions
gh copilot explain	gh copilot explain "curl -X POST ..."	Explains HTTP POST request

Command	Example Usage	What It Does
gh copilot explain	gh copilot explain "docker run -it ubuntu"	Explains Docker container launch
gh copilot explain	gh copilot explain "grep -r 'password' ."	Explains recursive search for sensitive keywords

GitHub Copilot VS Code Commands

These are used via Copilot Chat or inline suggestions in VS Code.

Command	Example Prompt	What It Does
/explain	Explain what this function does	Describes logic of selected code
/fix	Fix the bug in this loop	Suggests corrected code
/tests	Write unit tests for this function	Generates test cases
/optimize	Optimize this SQL query	Improves performance
/refactor	Refactor this into smaller functions	Breaks code into cleaner parts
/comment	Add comments to this class	Adds doc-style comments
/convert	Convert this Python code to JavaScript	Translates code between languages
/doc	Generate documentation for this module	Creates markdown or docstrings
/regex	Create regex to match email addresses	Generates regex pattern
/commit	Suggest a commit message for these changes	Proposes descriptive message
/review	Review this code for security issues	Flags vulnerabilities
/new	Create a new React component for login	Generates boilerplate
/clear	Clear chat history	Resets Copilot Chat

Command	Example Prompt	What It Does
@terminal	@terminal How do I install PostgreSQL?	Adds terminal context
@workspace	@workspace Refactor all functions using fetch()	Applies changes across files